

Payout Earnings Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Payout Earnings Sensitivity (PES), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PES achieves an annualized gross (net) Sharpe ratio of 0.30 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 29 (20) bps/month with a t-statistic of 3.19 (2.28), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities) is 20 bps/month with a t-statistic of 2.18.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should immediately reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This gradual incorporation of information manifests particularly in the relationship between corporate payout policies and earnings, where changes in the sensitivity of payouts to earnings may signal fundamental shifts in firm characteristics that investors only slowly recognize.

Despite extensive research on how investors process earnings information and payout announcements separately, the literature has largely overlooked how the joint dynamics of these two fundamental variables predict future returns. The sensitivity of payouts to earnings—how responsive dividend and repurchase policies are to earnings changes—potentially contains rich information about management’s confidence, investment opportunities, and financial constraints. Yet this signal may be complex and subtle enough that investors with limited attention fail to immediately incorporate it into prices, creating a systematic underreaction that generates predictable return patterns.

The slow diffusion of information provides a compelling framework for understanding why Payout Earnings Sensitivity (PES) predicts stock returns. When firms exhibit high sensitivity between their payout policies and earnings realizations, they reveal information about their fundamental characteristics that requires careful analysis to interpret. Following the limited attention framework of [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), investors facing cognitive constraints cannot process all available information simultaneously and must allocate their attention selectively. Complex signals embedded in the relationship between multiple financial

variables, such as the comovement of payouts and earnings, are particularly likely to be overlooked in favor of simpler, more salient information.

The gradual information diffusion mechanism documented by [Hong and Stein \(1999\)](#) and [Hong et al. \(2000\)](#) suggests that information travels slowly across the investor base, with sophisticated investors initially recognizing the signal’s importance before it gradually spreads to the broader market. PES represents precisely the type of nuanced information that would diffuse slowly—it requires analyzing the dynamic relationship between two variables over time rather than focusing on a single metric. This complexity means that even attentive investors may initially underreact to changes in payout-earnings sensitivity, as they must first recognize the pattern and then interpret its implications for future firm performance.

The underreaction should be most pronounced among stocks where information frictions are strongest. [Zhang \(2006\)](#) demonstrates that information uncertainty amplifies the market’s sluggish response to new information, while [Hou and Moskowitz \(2005\)](#) show that price delay is systematically related to firm characteristics associated with limited investor attention. For PES, we expect the return predictability to concentrate among smaller firms with lower analyst coverage and less institutional ownership, where the mechanisms that facilitate information diffusion operate least efficiently. These predictions align with the broader literature on limited attention, including [Peng and Xiong \(2006\)](#), which models how investors must allocate scarce attention across multiple information sources, naturally leading to delayed price discovery for complex signals.

Our empirical analysis reveals that Payout Earnings Sensitivity strongly predicts cross-sectional stock returns, consistent with gradual information diffusion. A value-weighted long/short trading strategy based on PES achieves an annualized gross Sharpe ratio of 0.30, with the high-PES portfolio outperforming the low-PES portfolio by 27 basis points per month (t-statistic = 2.35). The strategy’s performance

remains robust after controlling for known risk factors, generating a monthly alpha of 29 basis points (t -statistic = 3.19) relative to the Fama-French six-factor model. These results indicate economically significant return predictability that existing factor models cannot explain.

The evidence strongly supports the information diffusion hypothesis through the pattern of returns across size quintiles. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the PES strategy generates average returns of 14 basis points per month, with alphas ranging from 3 to 24 basis points depending on the factor model specification. While these large-cap returns are more modest than the overall strategy performance, their economic significance remains substantial given the greater liquidity and lower trading costs in this segment. The strategy maintains statistical significance with t -statistics reaching 2.25 for certain factor model specifications even within the large-cap universe.

Accounting for transaction costs using the composite bid-ask spread measure from [Chen and Velikov \(2022\)](#), the PES strategy continues to generate significant abnormal returns. The net Sharpe ratio of 0.25 places the strategy in the top 17% of anomalies in the factor zoo, while the monthly net alpha of 20 basis points relative to the six-factor model (t -statistic = 2.28) confirms that implementation costs do not eliminate the strategy’s profitability. Furthermore, when we control for the six most closely related anomalies alongside the six Fama-French factors, the strategy maintains an alpha of 20 basis points per month (t -statistic = 2.18), demonstrating that PES captures distinct information not subsumed by existing predictors.

This paper makes several contributions to the asset pricing literature on information diffusion and return predictability. While [Fama and French \(2001\)](#) and [Grullon and Michaely \(2002\)](#) examine payout policies and [Ball and Brown \(1968\)](#) and [Bernard and Thomas \(1989\)](#) study earnings announcement effects separately, we are the first to show that the sensitivity of payouts to earnings contains distinct predictive infor-

mation for returns. Our findings extend the gradual information diffusion literature by identifying a novel signal that emerges from the interaction of two fundamental accounting variables, demonstrating that investors’ limited attention extends to complex relationships between financial statement items.

Methodologically, we advance the literature by rigorously evaluating PES using the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), addressing concerns raised by [Harvey et al. \(2016\)](#) about data mining in anomaly discovery. Our analysis spans over 60 years of data and demonstrates that PES not only survives standard robustness tests but also adds incremental explanatory power beyond 154 existing anomalies from the factor zoo. The strategy’s ability to generate significant net alphas after transaction costs, documented using the high-frequency effective spread measures from [Chen and Velikov \(2022\)](#), establishes its practical relevance for investors and distinguishes it from many recently proposed factors that fail to survive implementation costs.

Our results have broader implications for understanding how markets process complex information and the limits of arbitrage. The persistence of PES-based return predictability, even among large-cap stocks with substantial analyst coverage, suggests that information complexity creates barriers to efficient pricing that extend beyond simple neglect of obscure firms. This finding contributes to the growing literature on the role of investor sophistication and attention constraints in generating asset pricing anomalies, as discussed in [Stambaugh et al. \(2012\)](#) and [McLean and Pontiff \(2016\)](#). By documenting how the joint dynamics of fundamental accounting variables predict returns through the slow diffusion channel, we provide new evidence on the specific types of information that markets struggle to efficiently incorporate into prices.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data necessary to construct the payout earnings sensitivity measure, spanning several decades of annual observations. We focus on U.S. equities traded on major exchanges, excluding financial firms and utilities due to their unique regulatory environments and accounting conventions. construction of our payout earnings sensitivity signal requires two key accounting variables. Operating income after depreciation (OIADP) represents a firm's operating performance after accounting for depreciation and amortization expenses, providing a measure of cash-generating ability from core operations before interest and taxes. Dividends (DVT) captures the total amount of common and preferred dividends declared on all equity capital during the fiscal year, representing the firm's cash distributions to shareholders. Both variables are measured in millions of dollars and collected at the fiscal year-end. construct the payout earnings sensitivity signal by computing the year-over-year change in operating income after depreciation scaled by lagged dividends: $(OIADP(t) - OIADP(t-1)) / DVT(t-1)$. This measure captures the sensitivity of a firm's earnings changes relative to its dividend commitment level, providing insight into how earnings volatility relates to the firm's payout policy. We calculate this signal annually using fiscal year-end values and require non-missing values for operating income in consecutive years and non-zero lagged dividends to ensure meaningful ratios. Firms with zero or missing dividend values in the denominator are excluded from the analysis, as are observations with extreme values in the top and bottom 1

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PES signal. Panel A plots the time-series of the mean, median, and interquartile range for PES. On average, the cross-sectional mean (median) PES is 4.00 (0.44) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PES data. The signal’s interquartile range spans -2.92 to 3.70. Panel B of Figure 1 plots the time-series of the coverage of the PES signal for the CRSP universe. On average, the PES signal is available for 3.66% of CRSP names, which on average make up 6.86% of total market capitalization.

4 Does PES predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PES using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PES portfolio and sells the low PES portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PES strategy earns an average return of 0.27% per month with a t-statistic of 2.35. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 1.60 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.79,

with a t-statistic of -12.57 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 323 stocks and an average market capitalization of at least \$858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -3-28bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PES trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in ten cases.

Table 3 provides direct tests for the role size plays in the PES strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PES, as well as average returns and alphas for long/short trading PES strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PES strategy achieves an average return of 14 bps/month with a t-statistic of 1.11. Among these large cap stocks, the alphas for the PES strategy relative to the five most common factor models range from 3 to 24 bps/month with t-statistics between 0.21 and 2.25.

5 How does PES perform relative to the zoo?

Figure 2 puts the performance of PES in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PES strategy falls in the distribution. The PES strategy’s gross (net) Sharpe ratio of 0.30 (0.25) is greater than 65% (83%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PES strategy (red line).² Ignoring trading costs, a \$1 invested in the PES strategy would have yielded \$3.47 which ranks the PES strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PES strategy would have yielded \$2.19 which ranks the PES strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PES relative to those. Panel A shows that the PES strategy gross alphas fall between the 36 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PES strategy has a positive net generalized alpha for five out of the five factor models. In these cases PES ranks between the 54 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PES add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PES with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PES or at least to weaken the power PES has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PES and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PES signal in these Fama-MacBeth regressions exceed 0.59, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on PES is 2.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the PES strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PES strategy earns alphas that range from 21-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.26, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PES trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.18.

7 Does PES add relative to the whole zoo?

Finally, we can ask how much adding PES to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PES signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PES grows to \$2576.57.

8 Conclusion

This study provides compelling evidence that Payout Earnings Sensitivity (PES) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PES-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PES delivers an annualized Sharpe ratio of 0.30 gross and 0.25 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. More importantly, the strategy generates statistically significant abnormal returns of 29 basis points per month (t-statistic = 3.19) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 20 basis point monthly alpha (t-statistic = 2.18) even after controlling for six closely related strategies—including growth measures, asset changes, and reversal effects—underscores that PES captures unique information about future returns not subsumed by existing factors.

ization on CRSP in the period for which PES is available.

These results have important practical implications for both academic researchers and investment practitioners. For academics, PES emerges as a distinct return predictor that enriches our understanding of the cross-sectional determinants of expected returns. For practitioners, the signal’s economic significance and robustness to transaction costs suggest it could be valuable for portfolio construction and risk management, particularly given its orthogonality to conventional factors.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodologies to address data mining concerns, the proliferation of documented anomalies raises questions about the true statistical significance of any individual signal. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of PES remains unexplored. Third, although we account for transaction costs, other implementation challenges such as market impact and capacity constraints may further erode real-world profitability.

Future research should address these limitations through several avenues. International evidence would strengthen claims about the pervasiveness of the PES effect and help distinguish between risk-based and behavioral explanations. Time-series analysis of the strategy’s performance across different market regimes could illuminate when and why PES predicts returns. Additionally, investigating the economic mechanisms underlying the predictive power of PES—whether through connections to firm fundamentals, investor behavior, or market frictions—would enhance our theoretical understanding. Finally, examining the signal’s performance in out-of-sample periods and its interaction with other documented anomalies could provide further insights into its practical applicability and contribution to the broader factor zoo.

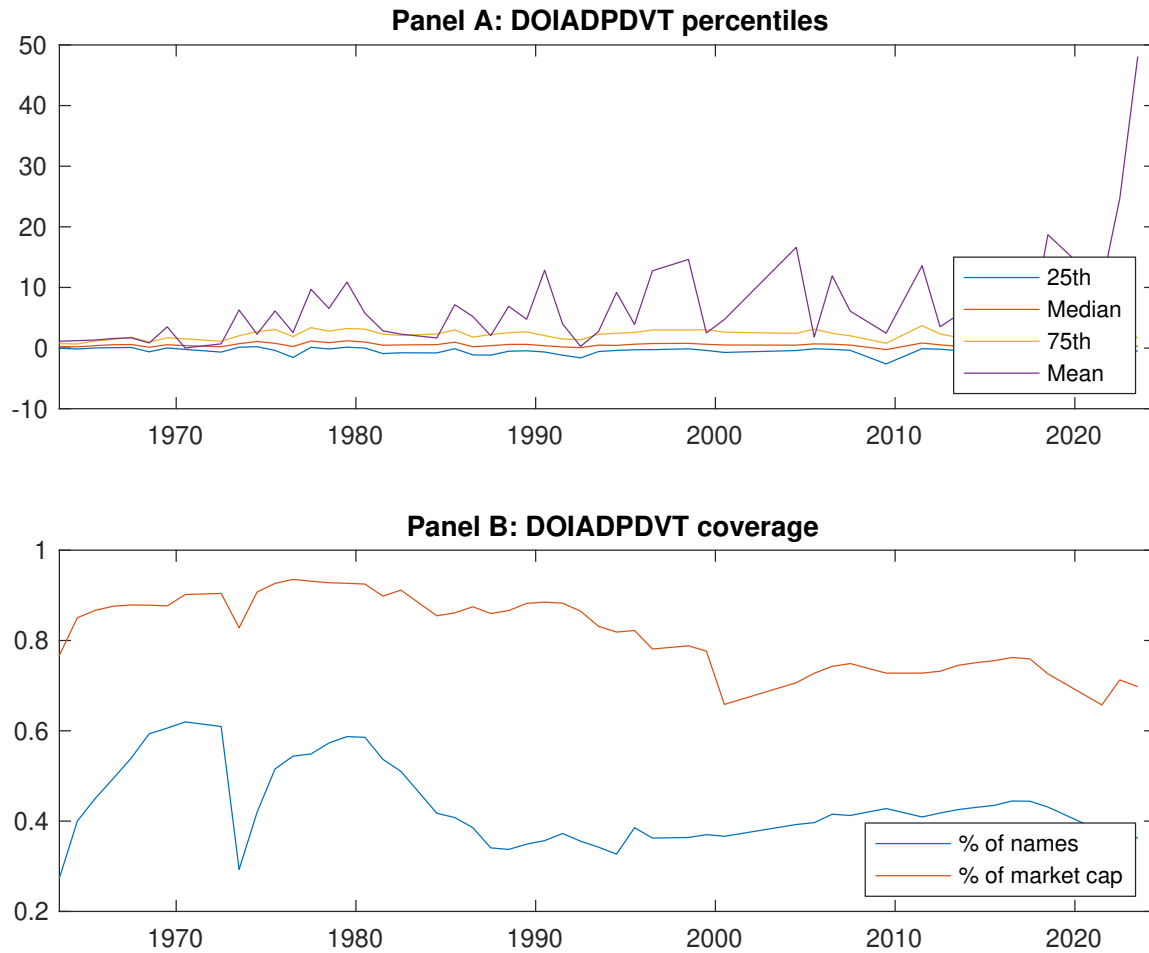


Figure 1: Times series of PES percentiles and coverage.
This figure plots descriptive statistics for PES. Panel A shows cross-sectional percentiles of PES over the sample. Panel B plots the monthly coverage of PES relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PES. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PES-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.32]	0.58 [3.83]	0.60 [4.08]	0.68 [4.05]	0.72 [3.40]	0.27 [2.35]
α_{CAPM}	-0.17 [-2.18]	0.10 [1.43]	0.12 [2.06]	0.11 [2.14]	0.01 [0.11]	0.18 [1.60]
α_{FF3}	-0.31 [-4.47]	0.01 [0.14]	0.08 [1.62]	0.09 [1.79]	0.02 [0.32]	0.33 [3.16]
α_{FF4}	-0.26 [-3.66]	-0.00 [-0.07]	0.07 [1.49]	0.09 [1.78]	0.05 [0.67]	0.31 [2.83]
α_{FF5}	-0.31 [-4.63]	-0.14 [-2.67]	-0.05 [-0.99]	0.01 [0.25]	0.02 [0.30]	0.33 [3.62]
α_{FF6}	-0.26 [-3.86]	-0.13 [-2.52]	-0.04 [-0.86]	0.02 [0.34]	0.03 [0.55]	0.29 [3.19]
Panel B: Fama and French (2018) 6-factor model loadings for PES-sorted portfolios						
β_{MKT}	1.09 [67.71]	0.95 [76.42]	0.91 [81.90]	1.00 [90.50]	1.14 [75.03]	0.05 [2.37]
β_{SMB}	0.08 [3.47]	-0.15 [-8.30]	-0.14 [-8.79]	-0.04 [-2.66]	0.14 [6.56]	0.06 [2.00]
β_{HML}	0.18 [5.75]	0.06 [2.39]	0.09 [4.33]	0.12 [5.82]	0.11 [3.76]	-0.07 [-1.61]
β_{RMW}	-0.17 [-5.36]	0.13 [5.42]	0.29 [13.38]	0.28 [12.96]	0.27 [8.93]	0.43 [10.07]
β_{CMA}	0.35 [7.63]	0.52 [14.69]	0.12 [3.87]	-0.10 [-3.32]	-0.44 [-10.15]	-0.79 [-12.57]
β_{UMD}	-0.07 [-4.60]	-0.01 [-0.69]	-0.01 [-0.67]	-0.01 [-0.57]	-0.02 [-1.49]	0.05 [2.32]
Panel C: Average number of firms (n) and market capitalization (me)						
n	424	329	323	358	434	
me (\$10 ⁶)	858	1725	1962	2021	1131	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PES strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [2.35]	0.18 [1.60]	0.33 [3.16]	0.31 [2.83]	0.33 [3.62]	0.29 [3.19]
Quintile	NYSE	EW	0.15 [2.10]	0.12 [1.61]	0.17 [2.43]	0.07 [0.95]	0.08 [1.19]	-0.01 [-0.19]
Quintile	Name	VW	0.33 [2.74]	0.25 [2.07]	0.39 [3.50]	0.35 [3.05]	0.37 [3.78]	0.32 [3.26]
Quintile	Cap	VW	0.15 [1.35]	0.04 [0.33]	0.19 [1.90]	0.20 [1.92]	0.28 [3.24]	0.27 [3.05]
Decile	NYSE	VW	0.31 [2.16]	0.23 [1.62]	0.38 [2.77]	0.32 [2.33]	0.33 [2.74]	0.28 [2.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [1.96]	0.12 [1.05]	0.26 [2.45]	0.24 [2.30]	0.22 [2.47]	0.20 [2.28]
Quintile	NYSE	EW	-0.03 [-0.41]					
Quintile	Name	VW	0.28 [2.33]	0.18 [1.52]	0.32 [2.81]	0.29 [2.59]	0.26 [2.71]	0.24 [2.47]
Quintile	Cap	VW	0.11 [1.01]		0.12 [1.24]	0.13 [1.29]	0.17 [1.98]	0.17 [1.93]
Decile	NYSE	VW	0.25 [1.74]	0.15 [1.07]	0.29 [2.11]	0.26 [1.88]	0.22 [1.82]	0.19 [1.58]

Table 3: Conditional sort on size and PES

This table presents results for conditional double sorts on size and PES. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PES. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PES and short stocks with low PES. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PES Quintiles					PES Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.61 [2.26]	1.00 [3.62]	0.87 [4.73]	0.94 [4.61]	0.84 [3.36]	0.23 [2.03]	0.24 [2.11]	0.26 [2.29]	0.16 [1.41]	0.13 [1.17]	0.05 [0.45]
	(2)	0.73 [3.18]	0.73 [3.98]	0.87 [4.93]	0.89 [4.34]	0.85 [3.49]	0.11 [1.17]	0.06 [0.59]	0.13 [1.37]	0.07 [0.69]	0.03 [0.36]	-0.02 [-0.20]
	(3)	0.64 [3.05]	0.66 [4.03]	0.82 [4.70]	0.87 [4.52]	0.87 [3.75]	0.22 [2.13]	0.15 [1.44]	0.22 [2.15]	0.15 [1.47]	0.17 [1.69]	0.11 [1.07]
	(4)	0.59 [2.95]	0.69 [4.24]	0.75 [4.56]	0.84 [4.49]	0.63 [2.85]	0.04 [0.40]	-0.03 [-0.30]	0.04 [0.38]	-0.04 [-0.36]	-0.07 [-0.71]	-0.14 [-1.42]
	(5)	0.46 [2.63]	0.56 [3.89]	0.55 [3.63]	0.59 [3.54]	0.60 [2.96]	0.14 [1.11]	0.03 [0.21]	0.17 [1.40]	0.19 [1.53]	0.24 [2.25]	0.24 [2.20]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PES Quintiles					PES Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	149	152	154	153	152	11	13	14	14	14	
	(2)	63	63	63	64	63	26	28	28	28	28	
	(3)	54	54	54	54	54	55	56	58	58	56	
	(4)	51	51	51	51	51	137	139	140	142	136	
(5)	53	54	53	54	53	1145	1425	1408	1404	1132		

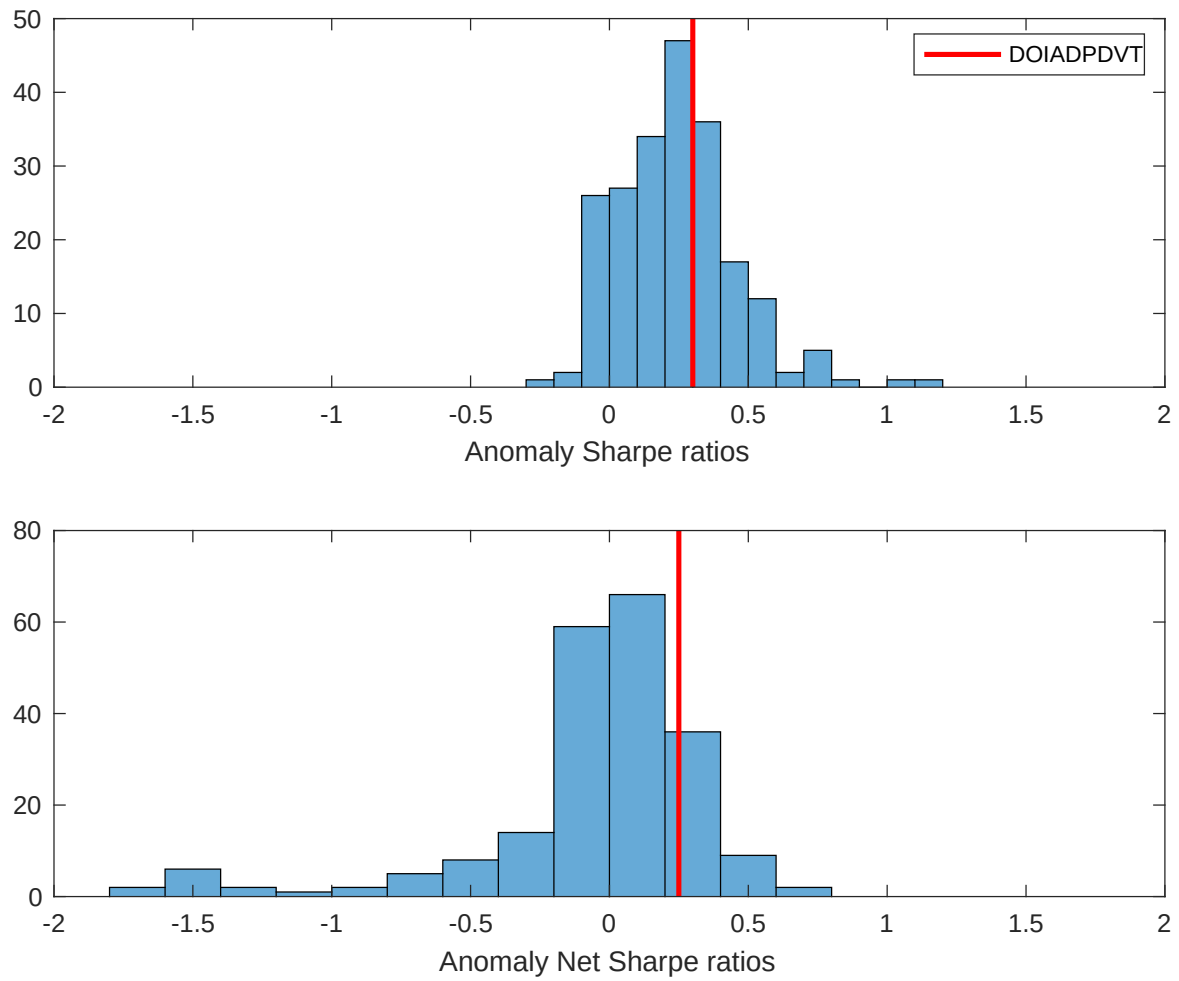


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PES with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

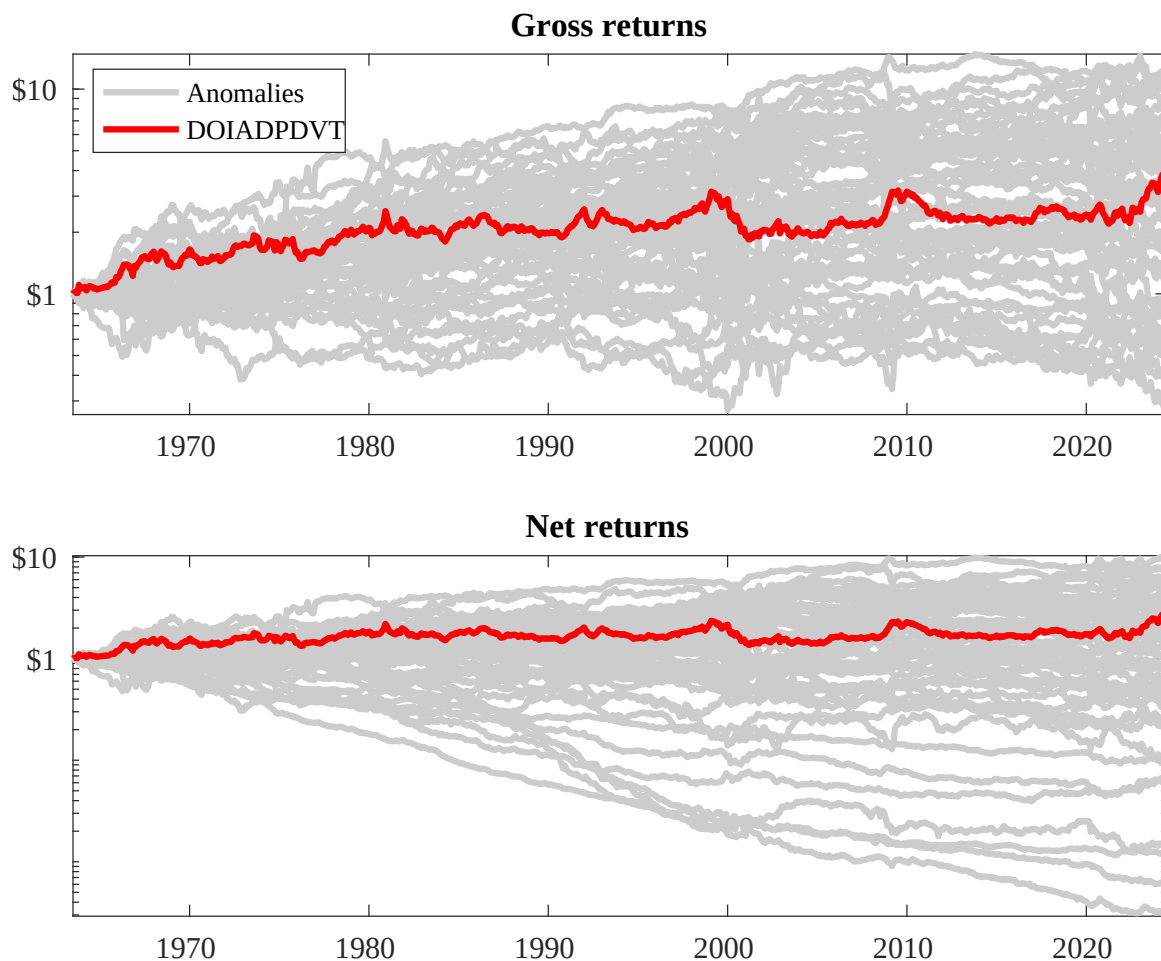
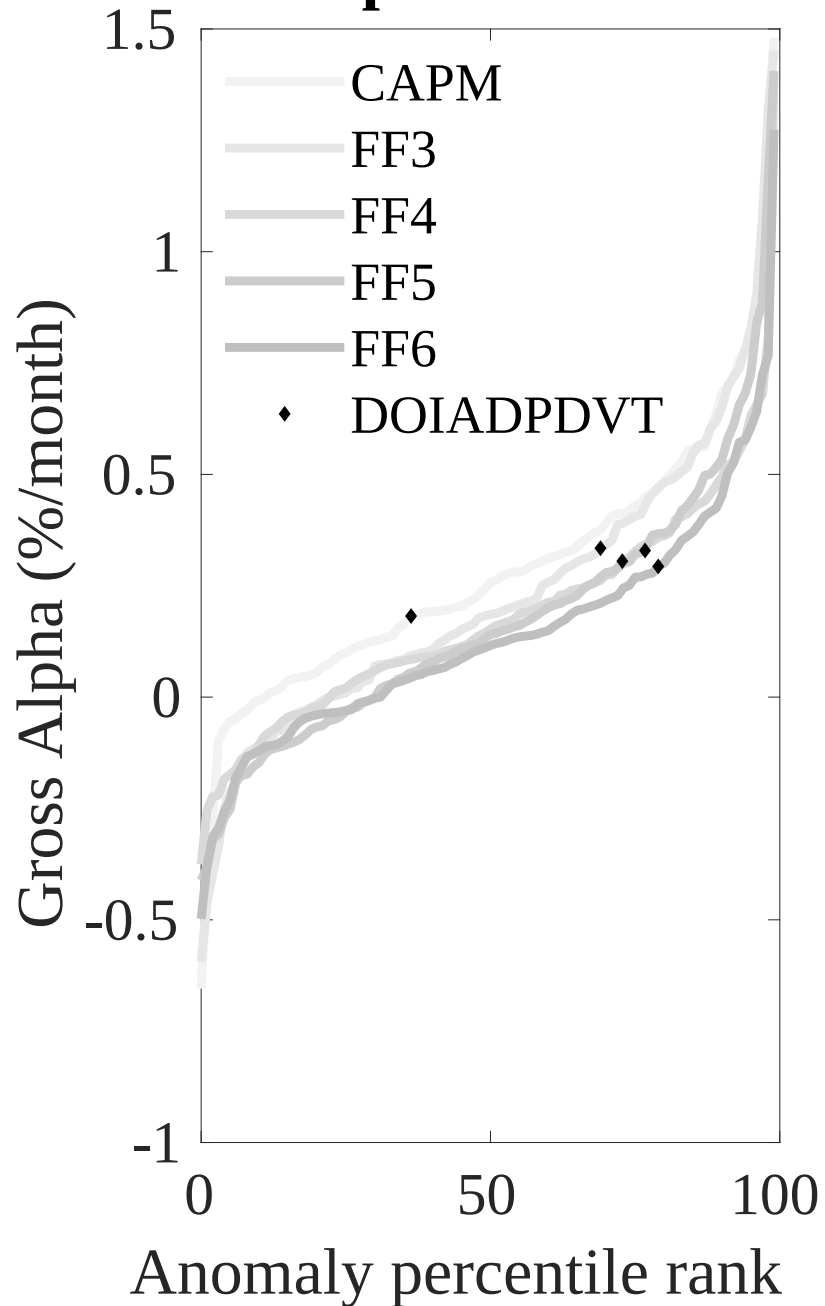


Figure 3: Dollar invested.

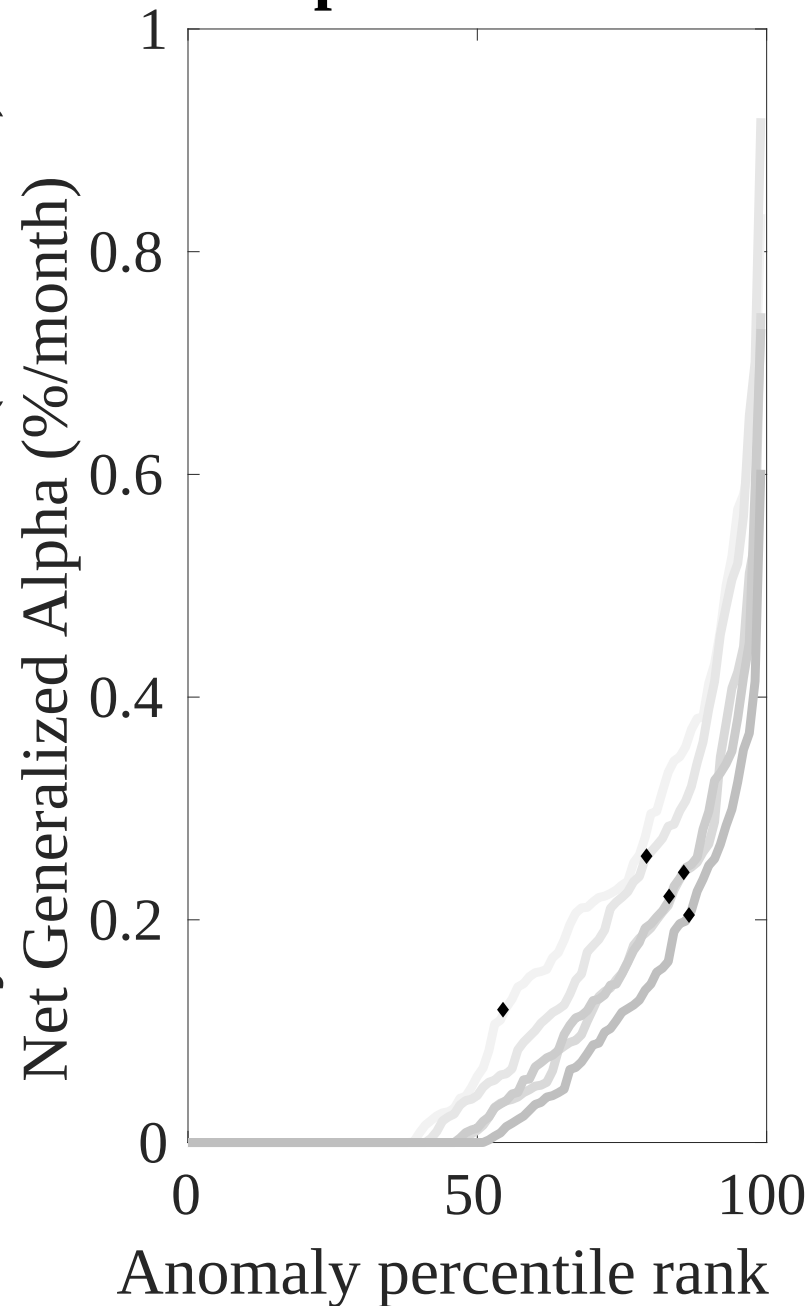
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PES trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



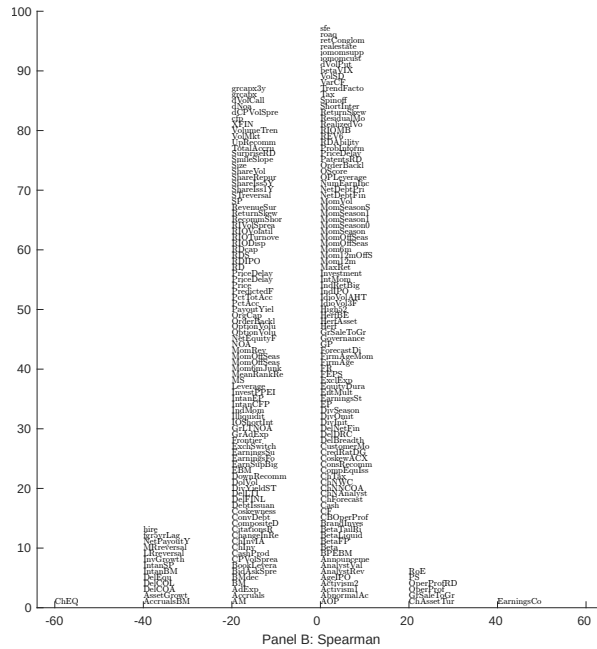
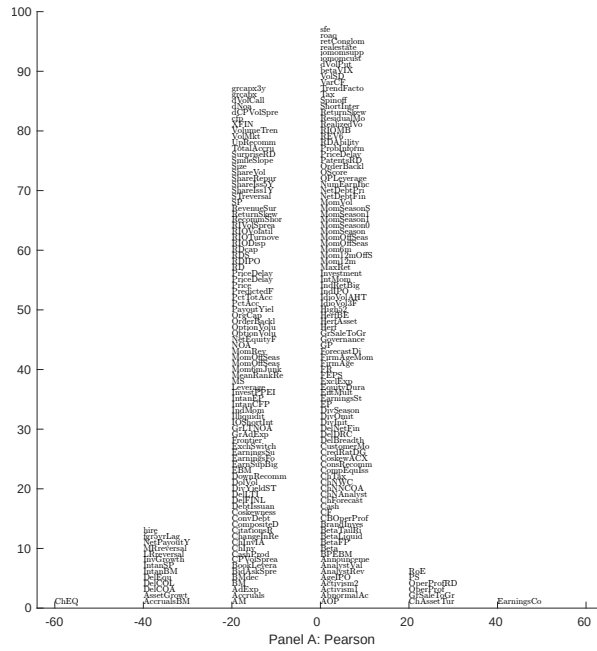


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PES. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

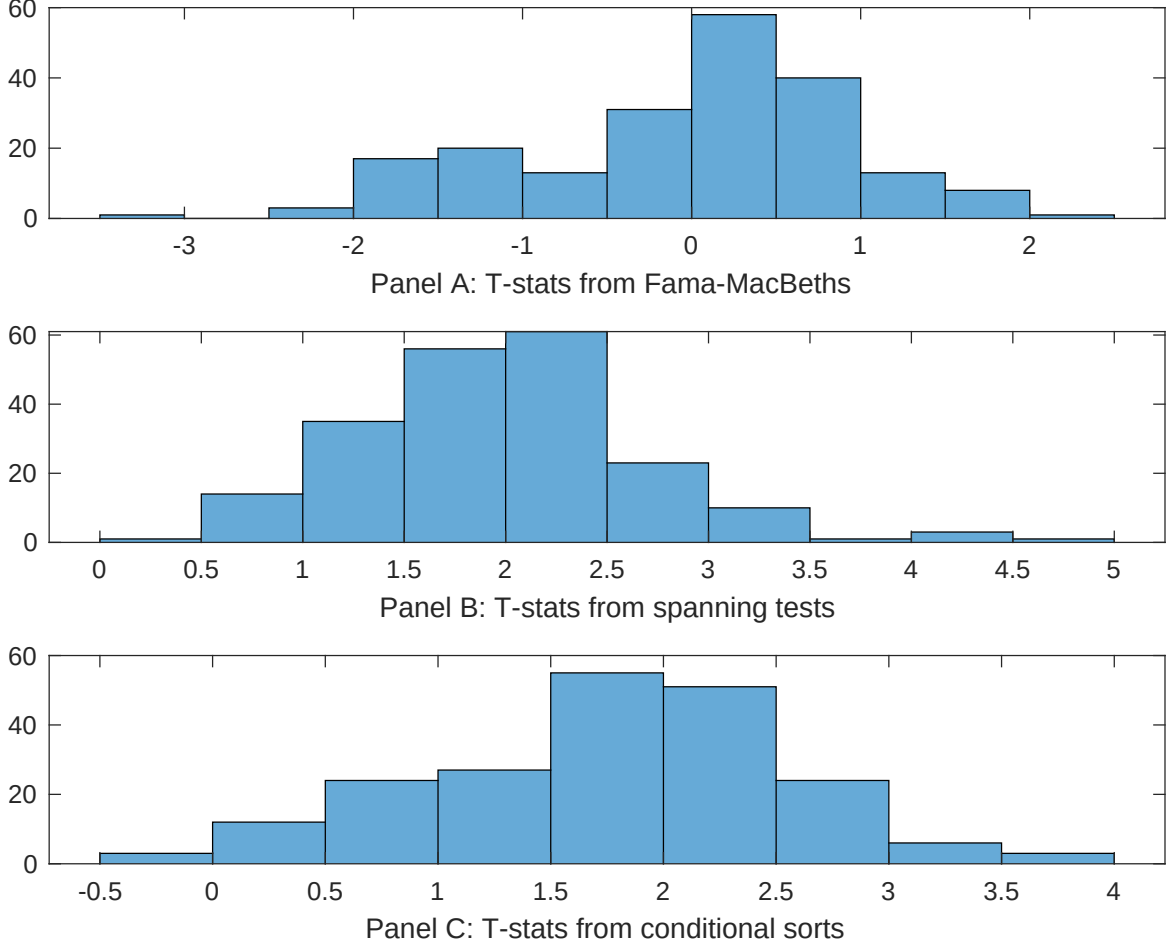


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PES. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [7.30]	0.12 [6.25]	0.13 [6.81]	0.12 [6.10]	0.12 [6.38]	0.12 [6.23]	0.12 [5.69]
PES	0.48 [1.80]	0.49 [1.85]	0.36 [1.47]	0.18 [0.72]	0.33 [1.34]	0.14 [0.59]	0.58 [2.14]
Anomaly 1	0.43 [3.59]						0.58 [0.40]
Anomaly 2		0.14 [3.60]					0.29 [0.60]
Anomaly 3			0.84 [7.11]				0.59 [4.76]
Anomaly 4				0.70 [4.80]			0.15 [1.10]
Anomaly 5					0.15 [1.62]		0.50 [0.56]
Anomaly 6						0.88 [1.86]	-0.30 [-0.59]
# months	720	720	732	713	727	732	708
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PES trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PES} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29 [3.25]	0.28 [3.06]	0.28 [3.03]	0.21 [2.26]	0.26 [2.90]	0.23 [2.53]	0.20 [2.18]
Anomaly 1	-27.56 [-5.64]						-44.05 [-6.09]
Anomaly 2		-5.16 [-1.13]					27.58 [4.13]
Anomaly 3			-19.20 [-3.53]				-2.95 [-0.44]
Anomaly 4				-20.76 [-4.13]			-13.62 [-2.67]
Anomaly 5					-13.76 [-5.21]		-9.28 [-3.44]
Anomaly 6						-14.77 [-3.42]	-11.48 [-2.42]
mkt	4.13 [1.92]	5.13 [2.34]	5.01 [2.30]	4.54 [2.07]	5.65 [2.62]	5.57 [2.55]	3.51 [1.65]
smb	6.28 [2.02]	5.43 [1.71]	6.90 [2.17]	4.60 [1.45]	11.33 [3.45]	4.93 [1.56]	10.04 [3.10]
hml	-4.64 [-1.11]	-7.38 [-1.72]	-7.38 [-1.75]	-5.04 [-1.17]	-3.36 [-0.79]	-2.36 [-0.52]	3.06 [0.68]
rmw	42.34 [10.09]	43.17 [10.04]	43.93 [10.34]	42.99 [10.08]	37.77 [8.75]	44.82 [10.52]	40.32 [9.44]
cma	-50.77 [-6.58]	-72.28 [-9.33]	-54.29 [-6.02]	-58.02 [-7.60]	-66.81 [-10.47]	-69.20 [-10.42]	-31.78 [-3.35]
umd	5.36 [2.51]	4.73 [2.17]	4.27 [1.97]	5.70 [2.61]	4.07 [1.91]	3.90 [1.79]	5.44 [2.56]
# months	732	732	732	713	728	732	709
$\bar{R}^2(\%)$	46	44	45	45	46	45	50

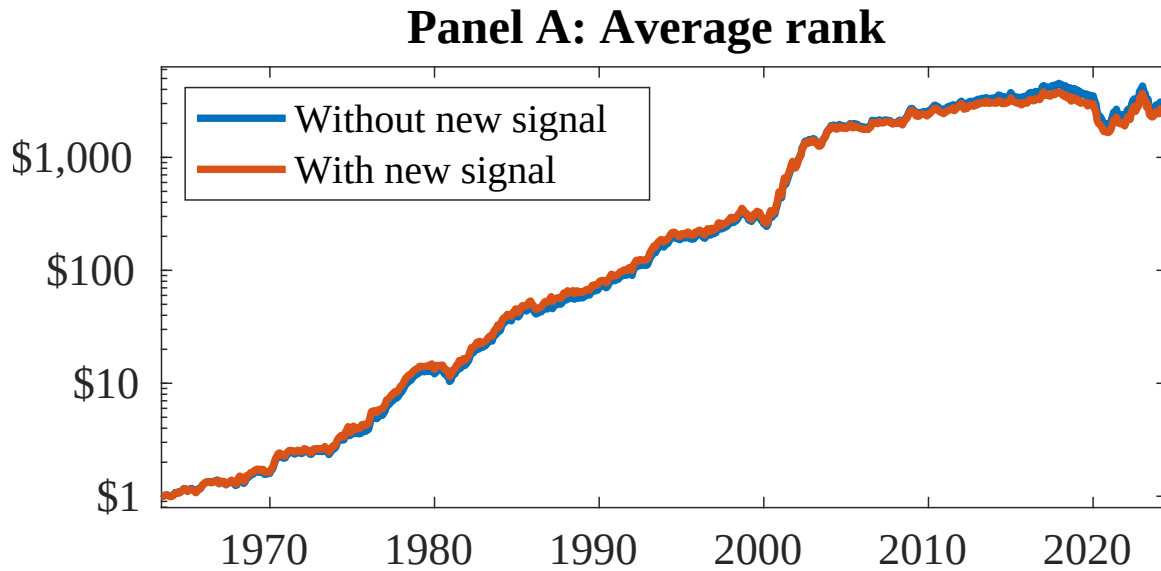


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PES. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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